

Network Configuration

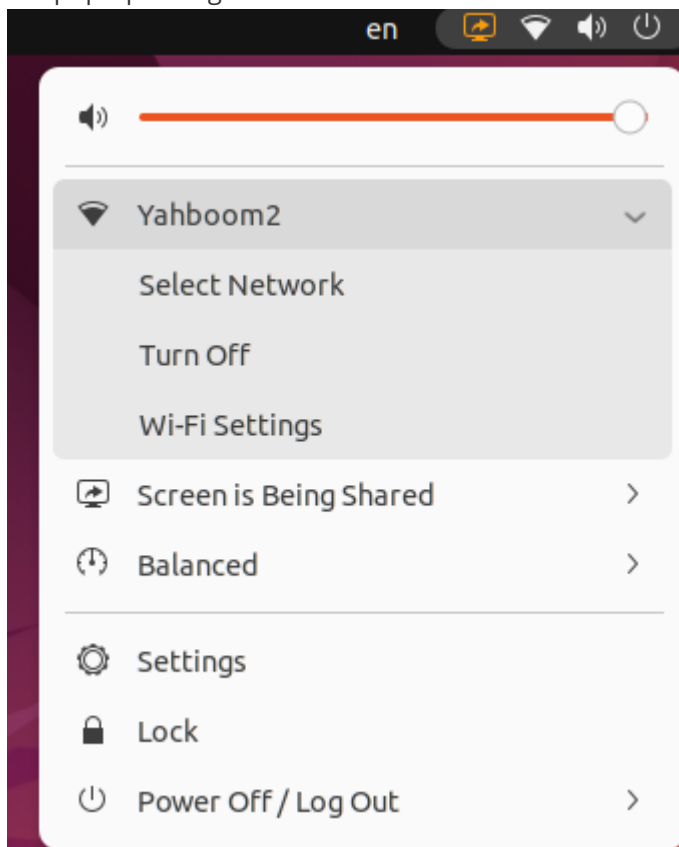
Network Configuration

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1. Wireless Network

• Desktop System

Connect to Wi-Fi using the Wi-Fi management tool in the upper right corner of the menu bar, as shown in the figure below. Click the target Wi-Fi name, then enter the Wi-Fi password in the pop-up dialog box.



• Command Line

Wireless network configuration can be done via the command line, following the steps below:

```
sudo nmcli device wifi rescan # Scan for Wi-Fi networks
sudo nmcli device wifi list # List detected Wi-Fi networks
```

Run the following command to connect to the specified Wi-Fi network.

```
sudo wifi_connect "SSID" "PASSWD"
```

The message "successfully activated" in the terminal indicates a successful Wi-Fi connection.

If a message indicating the access point cannot be found appears after attempting to connect, run the command `sudo nmcli device wifi rescan` to rescan for Wi-Fi networks and retry the connection.

```
root@ubuntu:~# sudo wifi_connect "WiFi-Test" "12345678"
Error: No network with SSID 'WiFi-Test' found.
```

2. Wired Network

Network configuration via Netplan has only been validated on `RDK S100` / `RDK S600` and is not supported on other platforms temporarily.

- The `RDK S600` root file system is built on Ubuntu-24.04 and does not support enabling or disabling network interfaces using `ifup/ifdown` by default.
- **Static IP and Subnet Mask:** To set a static IP address and subnet mask for a network interface, use the `addresses` field with the CIDR notation for the subnet mask.
- **DHCP Configuration:** To enable automatic IP address assignment via DHCP (Dynamic Host Configuration Protocol) for a specified network interface, set the `dhcp4` or `dhcp6` field value to `yes`.
- **Custom MAC Address:** Use the `macaddress` field to assign a custom MAC address to the network interface.
- **Custom DNS Address:** Use the `nameservers` field to specify custom DNS addresses for the network interface.

An example of network configuration using Netplan is shown below:

```
sudo vim /etc/netplan/99-hobot-net.yaml
```

```
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    eth0:
      dhcp4: yes
      dhcp6: yes
      nameservers:
        addresses: [10.9.1.2, 8.8.8.8, 8.8.4.4]
    eth1:
      addresses:
        - 192.168.127.10/24
      nameservers:
        addresses: [10.9.1.2, 8.8.8.8, 8.8.4.4]
```

After making changes, run `sudo netplan apply` to apply the configuration.

3. DNS Service

DNS (Domain Name Server) is a server that translates domain names into their corresponding IP addresses.

The DNS configuration of the development board is managed through the `/etc/systemd/resolved.conf` file. Users can modify this file to complete DNS-related configurations as follows:

1. Edit the `resolved.conf` file and add the DNS server address, for example:

```
DNS=8.8.8.8 114.114.114.114
```

2. Apply the DNS configuration using the following commands:

```
sudo systemctl restart systemd-resolved
sudo systemctl enable systemd-resolved
sudo mv /etc/resolv.conf /etc/resolv.conf.bak
sudo ln -s /run/systemd/resolve/resolv.conf /etc/
```

4.Proxy Configuration

Proxy configuration refers to setting up a network proxy. In network communication, a proxy server acts as an intermediary between the client and the target server. The client's request is first sent to the proxy server, which then forwards it to the target server. The target server's response is also returned to the client through the proxy server.

Edit the `~/.bashrc` or `/etc/environment` file. If configuring the proxy for the current user, edit `~/.bashrc`; if configuring the proxy for all users, edit `/etc/environment`.

Add the following content to the file (using HTTP proxy as an example):

```
http_proxy=http://proxy_server_address:port
https_proxy=http://proxy_server_address:port
ftp_proxy=http://proxy_server_address:port
no_proxy=localhost,127.0.0.1
```

After saving the file, run the following command to apply the configuration:

```
source ~/.bashrc
```